

2026-05-13

Idea

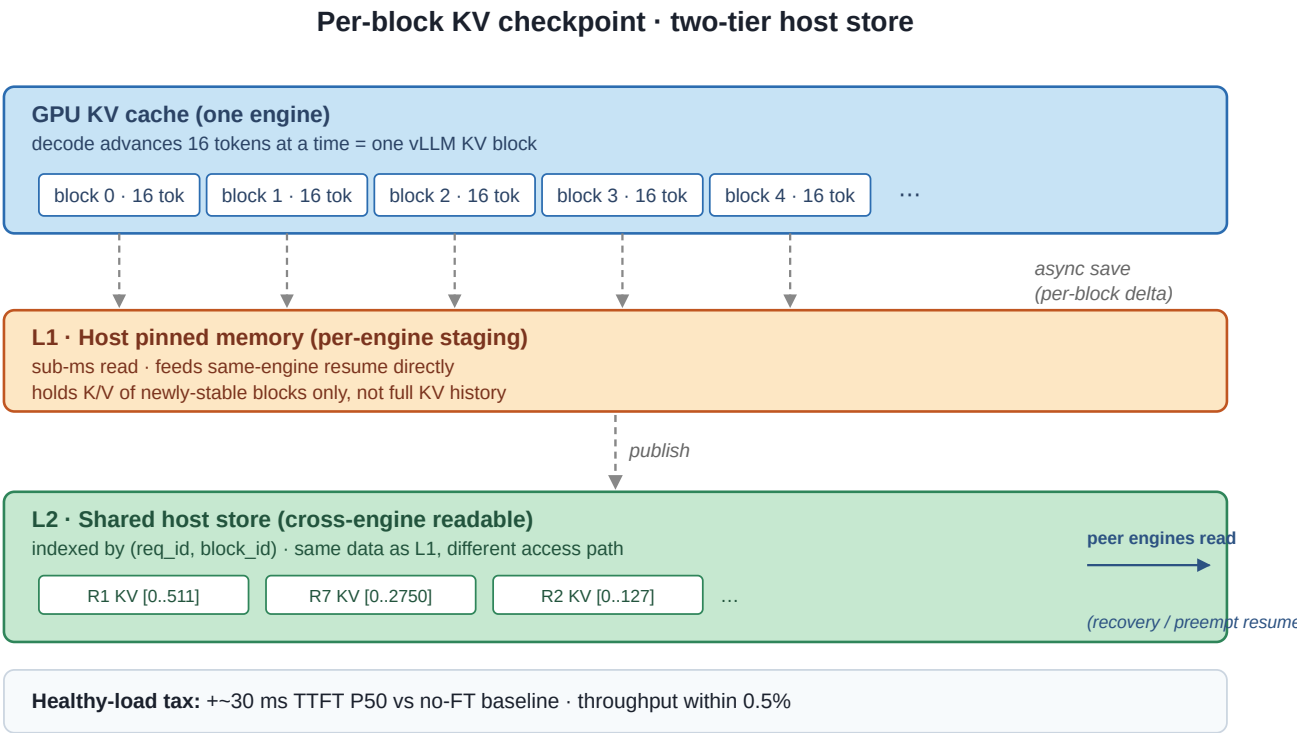
- **Old idea:** host-side KV checkpoint for fault tolerance — when an engine dies, a peer resumes from the checkpoint. Problem: failure probability is too low; the benefit doesn't carry a paper.
- **New idea:** leverage the KV checkpoint mechanism to make preemption cheap, so the scheduler can better meet SLO requirements and improve overall goodput. **We reuse the resume-without-reprell path from the old idea** — the rest of the design (slack-based preempt policy) is new.

Design

KV checkpoint mechanism

What	per-block delta (only newly-stable blocks)
When	every 16 tokens decoded (one vLLM KV block)
Where	L1 host pinned (same-engine) + L2 shared host store (cross-engine readable)

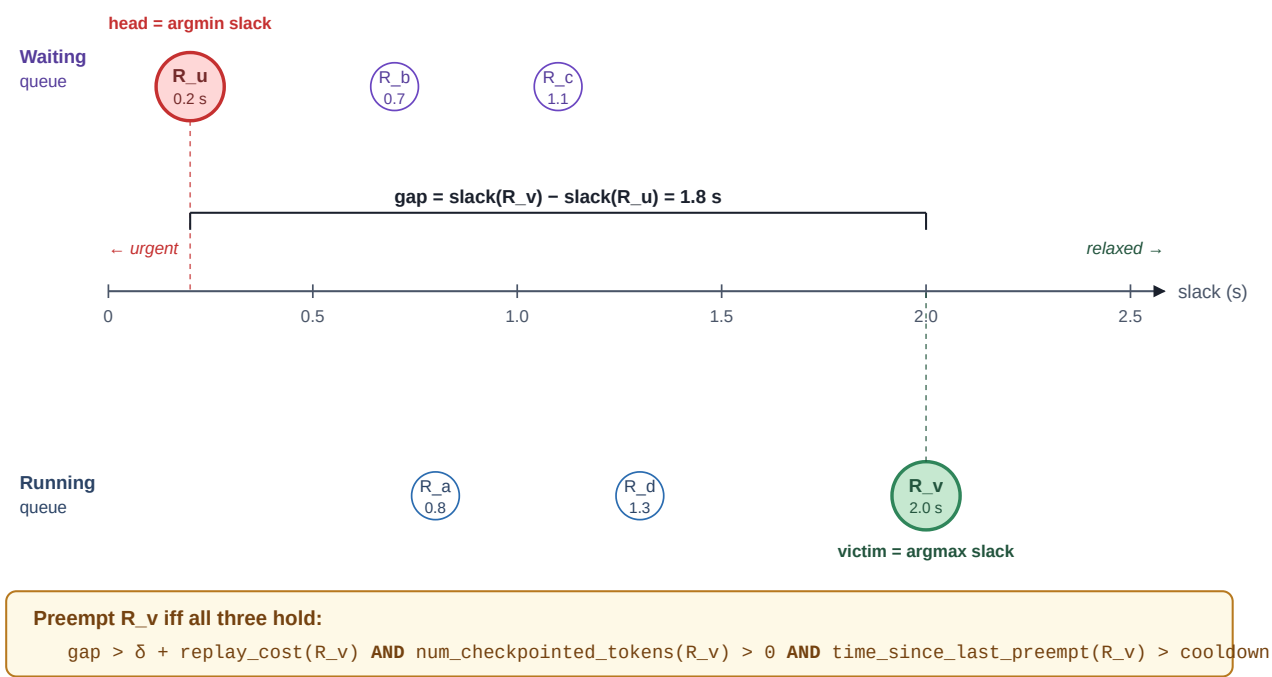
Overhead under normal load is small: TTFT P50 goes up by about 30 ms, throughput stays within 0.5%.



Slack-based preempt picker

- **slack** = wall-clock remaining until the request misses its SLO
- Each scheduler step: pick max-slack from running queue (victim), min-slack from waiting queue (head)
- Preempt iff (three AND): $gap > \delta + replay_cost \cdot victim\ has\ checkpoint \cdot cooldown\ elapsed$

Slack picker · pick laziest tail, admit most urgent head



Method

Setup

Hardware	2× A6000 (+ L40S portability)
Model	Qwen2.5-7B-Instruct fp16, <code>max_model_len 32K</code>
Workloads	RULER 16K (target) · ShareGPT (don't break)
Arrivals	Poisson · 3-tier QoS (tight / normal / loose)
SLO tiers	baseline P95 × {2×, 3×, 6×}
Baselines	<code>vllm_fcfs</code> · <code>reroute_no_ckpt</code> · <code>ours_no_picker</code> · <code>ours</code>
Seeds	3 per config

Experiments

	Measures	Status
E_M1	SLO attainment vs QPS	A6000 done · L40S in progress
E_M4	picker ablation (+ <code>ours_no_picker</code>)	scheduled this week
E_D1	failover gap (FT use case, bonus)	RULER 16K demo working

Submission

- **Target:** APSys 2026 Workshop

- **Deadline:** May 20 (Wed)